

Operator-Spectral Truncated Priors for Query-Efficient Biomedical QAOA Graph Optimization

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Abstract

Low-depth QAOA is often limited less by circuit definition than by the number of objective queries available for tuning angles on each graph. Recent spectral-truncation kernels show that finite-rank, noncommutative C^* -algebraic products can bridge separable and commutative operator-valued kernels while retaining tractable computation. This paper turns that idea into a different object: an executable *variational-search policy* for graph optimization. We introduce OST-QAOA, which maps each graph to a truncated operator on the $2p$ -dimensional QAOA angle space, uses the commutator of two graph-derived operators to encode noncommuting local/global structure, and converts the resulting low-rank covariance into query-ranked search directions. We prove that the truncated prior is positive definite and Loewner-contractive, that it reduces to the commutative diagonal prior in the limit of a single retained direction, and that the expected number of objective queries to reach a target ratio scales with the prior’s effective dimension $d_{\text{eff}}(n)$, which the truncation minimizes—the query-budget analogue of the representation-versus-complexity tradeoff of spectral-truncation kernels. The method is implemented as the installable package `uq-qaoa`; changing depth, query budget, truncation rank, commutator weight, seed, or graph families regenerates every figure, table, CSV, and query trace in this manuscript. In exact-statevector MaxCut experiments at $p = 3$, $Q = 24$, rank 4, commutator weight 4.0, and seed 260424803, OST-QAOA reaches 0.818 ± 0.018 mean approximation ratio across 16 held-out graphs, improving over a budget-matched TQA coordinate-refinement baseline by $+0.110$ (paired bootstrap 95% interval $[+0.080, +0.141]$, 16/16 instances, sign-test $p < 10^{-4}$), and reaching 98% of the best observed ratio in 11.6 queries against 25.0 for the baseline—a $2.2\times$ query reduction. A truncation sweep exhibits the predicted interior optimum: under-truncation and no truncation both degrade. Restricting the search to diagonal (commutative) directions removes a measurable part of the advantage (0.806 ± 0.020), localizing the gain to the off-diagonal operator geometry. The contribution is a reproducible operator-valued prior, with theory, that connects spectral truncation, graph structure, and query-efficient quantum variational search.

1 Introduction

Biomedical graph optimization problems such as pathway subset selection, patient-similarity partitioning, drug-combination screening, and molecular interaction pruning are naturally expressed as graph objectives. QAOA provides a variational quantum template for such objectives, but practical use requires a classical policy that spends a small number of expensive objective queries well. A warm-start point alone is not enough: the optimizer also needs to know which angle directions deserve exploration under a fixed query budget.

The central idea of this work is to learn and search in *operator space* rather than only in parameter-vector space. For a graph G , OST-QAOA constructs two $2p \times 2p$ operators from the graph Laplacian spectrum, degree moments, and topology features. Their noncommuting

product defines a commutator energy, and a truncated spectral decomposition of the resulting angle operator gives a low-rank covariance. The covariance is not used as a passive uncertainty estimate; it becomes the ranked list of search directions evaluated by the query policy.

This is motivated by, but distinct from, the spectral-truncation kernel framework of Hashimoto et al.⁵. That paper introduces positive-definite C^* -algebraic kernels whose truncation parameter controls noncommutativity and balances representation power against model complexity. Here the output is not a supervised vector-valued regression kernel. It is a graph-conditioned posterior over QAOA angles, together with a deterministic objective-query policy that can be audited line by line.

The contributions are:

1. **Operator-spectral QAOA priors.** OST-QAOA constructs a truncated, noncommuting operator on QAOA angle space—not a scalar graph feature or a diagonal uncertainty vector—from two graph-derived generators $\mathcal{A}_G, \mathcal{B}_G$ that do not commute, and spectrally truncates it to a rank- n covariance.
2. **Noncommutative query geometry.** The off-diagonal structure of the operator covariance is used operationally: the query policy probes the *collective* eigendirections of Σ_G rather than the raw angle coordinates. Restricting the search to the diagonal (commutative) directions removes a measurable part of the advantage (table 3).
3. **A query-budget truncation tradeoff with theory.** We prove that the truncated prior is positive definite and Loewner-contractive, that it reduces to the commutative diagonal prior in the $n \rightarrow 1$ limit, and that the expected number of objective queries to reach a target ratio scales with the prior’s effective dimension $d_{\text{eff}}(n)$ (section 3.4). This is the query-budget analogue of the representation-versus-complexity tradeoff of spectral-truncation kernels⁵, and a truncation sweep exhibits the predicted interior optimum n^* (fig. 3 and table 4).
4. **Reproducible, installable artifact.** Every figure, table, CSV, and query trace is regenerated by the package at $p = 3$, $Q = 24$, rank 4, commutator weight 4.0 from a single seed. The package installs with `pip install .` from `submission/code` and exposes both a CLI (`uqqaoa-reproduce`) and a Python API for retuning every experimental parameter.

2 Problem Setting

Let $G = (V, E)$ be an undirected graph with $n = |V|$. The MaxCut Hamiltonian is

$$H_C(G) = \sum_{(i,j) \in E} \frac{1 - Z_i Z_j}{2}. \quad (1)$$

At depth p , the QAOA parameter vector is $\theta = (\gamma_1, \dots, \gamma_p, \beta_1, \dots, \beta_p) \in [0, \pi]^{2p}$. The state and objective are

$$|\psi_G(\theta)\rangle = \prod_{\ell=1}^p e^{-i\beta_\ell H_M} e^{-i\gamma_\ell H_C(G)} |+\rangle^{\otimes n}, \quad f_G(\theta) = \langle \psi_G(\theta) | H_C(G) | \psi_G(\theta) \rangle. \quad (2)$$

We report the approximation ratio $r_G(\theta) = f_G(\theta)/C_G^*$, where C_G^* is computed exactly by brute force for the small audit graphs used here.

An algorithm receives a fixed query budget Q , where one query is one exact evaluation of $f_G(\theta)$. Each method is measured by the best ratio found within the same Q . This query model is deliberately simple: it isolates the angle-search policy from finite-shot readout and hardware noise.

3 Method

3.1 Graph-to-operator map

For graph G , let λ_i denote normalized Laplacian eigenvalues and let d_i denote centered degree values. Define two symmetric angle-space operators,

$$\mathcal{A}_G = \text{Toeplitz}(0.70 m_\lambda + 0.30 \phi(G)), \quad (3)$$

$$\mathcal{B}_G = \text{sym}[\text{roll Toeplitz}(0.55 m_d + 0.45 \phi(G)_{\text{rev}})], \quad (4)$$

where m_λ and m_d are spectral and degree moments of length $2p$, and $\phi(G)$ is the normalized graph descriptor used by the package. The noncommutative interaction is

$$\mathcal{C}_G = [\mathcal{A}_G, \mathcal{B}_G] = \mathcal{A}_G \mathcal{B}_G - \mathcal{B}_G \mathcal{A}_G. \quad (5)$$

The final raw angle operator is

$$\mathcal{O}_G = \text{sym}\left(0.55 \mathcal{A}_G + 0.45 \mathcal{B}_G + \omega \mathcal{C}_G^\top \mathcal{C}_G\right), \quad (6)$$

with $\omega = 4.0$ in the manuscript run.

Definition 1 (Spectral truncation on angle space). *Let $\mathcal{O}_G = U \Lambda U^\top$. For truncation rank r , define*

$$\mathcal{O}_{G,r} = U_r \Lambda_r U_r^\top, \quad (7)$$

where U_r contains the r leading eigenvectors after nonnegative clipping of the corresponding eigenvalues.

The truncation parameter r is analogous in spirit to spectral truncation in C^* -algebraic kernels, but here it controls the number of QAOA angle directions that can be emphasized by the search covariance.

3.2 Posterior from a training library

The package builds a deterministic offline library of small training graphs. For each training graph, a budgeted reference optimizer records a high-quality angle vector. For a test graph G , the vectorized upper triangle of $\mathcal{O}_{G,r}$, graph features, commutator norm, and effective rank form the operator signature $s(G)$. The k nearest library entries receive Gaussian weights

$$w_i(G) \propto \exp\left[-\frac{\|z(G) - z(G_i)\|_2^2}{2h_G^2}\right], \quad (8)$$

where $z(\cdot)$ is the standardized signature and h_G is the median distance among the selected neighbours. The prior mean blends the weighted library mean with the fixed TQA schedule:

$$\mu_G = (1 - \alpha) \sum_i w_i(G) \theta_i + \alpha \theta_{\text{TQA}}, \quad \alpha = 0.25. \quad (9)$$

The covariance is operator-dominated: the truncated operator supplies the leading search geometry and the neighbour scatter $\Sigma_G^{\text{nbr}} = \sum_i w_i(G) (\theta_i - \mu_G)(\theta_i - \mu_G)^\top$ a data-driven floor,

$$\Sigma_G = \beta \frac{\mathcal{O}_{G,r}}{\text{tr}(\mathcal{O}_{G,r})} + \left(1 - \frac{\beta}{2}\right) \frac{\Sigma_G^{\text{nbr}}}{\text{tr}(\Sigma_G^{\text{nbr}})} + \epsilon I, \quad \beta = 0.8, \epsilon = 0.025, \quad (10)$$

so that the spectral-truncation parameter r and the noncommuting operator geometry, rather than the diagonal neighbour variances, set the directions the query policy probes.

3.3 Query policy

The search evaluates a small set of anchors and then probes the leading eigendirections of Σ_G . The anchors are TQA, the operator posterior mean, the nearest operator-neighbour angle, and the global midpoint prior. The remaining queries evaluate positive and negative perturbations around the incumbent best angle, with step sizes proportional to the square root of the directional variance.

Query-policy steps. The executable policy is:

1. build $\mathcal{O}_{G,r}$ from eq. (6) and compute (μ_G, Σ_G) from eq. (10);
2. evaluate the anchors $\{\theta_{\text{TQA}}, \mu_G, \theta_{\text{nearest}}, \theta_{\text{global}}\}$;
3. diagonalize $\Sigma_G = V\Lambda V^\top$;
4. for scales $s \in \{1.0, 0.55, 0.28\}$, evaluate incumbent perturbations $\pm s\sqrt{\lambda_j}v_j$ in descending variance order until the query budget is exhausted;
5. return the best angle and full query trace.

3.4 Theoretical properties

We record properties of the operator prior (μ_G, Σ_G) and its truncation. They isolate what the construction contributes—a valid, contractive search frame whose noncommutative truncation controls query cost—and make precise the sense in which the commutative diagonal prior is a degenerate special case. Proofs are short and given inline.

Proposition 1 (Positive definiteness and a valid search frame). *For any floor $\epsilon > 0$, commutator weight $\omega \geq 0$, and operator weight $\beta \geq 0$, the matrix Σ_G in eq. (10) is symmetric positive definite. Hence its eigendecomposition $\Sigma_G = V\Lambda V^\top$ yields an orthonormal frame V with strictly positive directional variances λ_j , so the query policy of section 3.3 is well defined.*

Proof. The neighbour scatter is a nonnegative combination of rank-one PSD terms and is PSD; the truncated operator $\mathcal{O}_{G,r} = U_r \Lambda_r U_r^\top$ is PSD by the nonnegative clipping of definition 1; and $\epsilon I \succ 0$. A sum of PSD matrices with a strictly positive-definite summand is positive definite; symmetry is explicit. $\square$

Proposition 2 (Monotone spectral truncation). *Order the clipped eigenvalues of $\mathcal{O}_G$ as $\sigma_1 \geq \dots \geq \sigma_{2p} \geq 0$. Then truncation is Loewner-monotone in the rank, $\mathcal{O}_{G,r} \preceq \mathcal{O}_{G,r+1}$ for every r , with $\text{tr } \mathcal{O}_{G,r} = \sum_{j \leq r} \sigma_j$ and $\mathcal{O}_{G,2p}$ the PSD part of $\mathcal{O}_G$. The truncation parameter therefore interpolates between a one-direction prior ($r=1$) and the full operator prior ($r=2p$).*

Proof. $\mathcal{O}_{G,r+1} - \mathcal{O}_{G,r} = \sigma_{r+1} u_{r+1} u_{r+1}^\top \succeq 0$ since $\sigma_{r+1} \geq 0$; the trace identity follows from orthonormality of the u_j . $\square$

Proposition 3 (Commutative limit recovers the diagonal prior). *The commutator weight ω parameterises noncommutativity: at $\omega = 0$ the raw operator $\mathcal{O}_G = \text{sym}(0.55 \mathcal{A}_G + 0.45 \mathcal{B}_G)$ is independent of the commutator $\mathcal{C}_G = [\mathcal{A}_G, \mathcal{B}_G]$; and restricting Σ_G to its diagonal makes the eigenframe V the coordinate axes, in which case section 3.3 reduces exactly to per-angle coordinate refinement. The commutator-off and diagonal-covariance variants of section 4 are therefore the commutative special cases of OST-QAOA, and any gap between them and the full method measures the operational value of the noncommutative geometry.*

Definition 2 (Effective search dimension). *For a PSD covariance Σ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$, the effective dimension is the participation ratio $d_{\text{eff}}(\Sigma) = (\sum_j \lambda_j)^2 / \sum_j \lambda_j^2 \in [1, d]$: the number of directions carrying comparable variance.*

Proposition 4 (Truncation reduces effective dimension). *In the operator-dominated regime $\beta \rightarrow 1$ of eq. (10) (up to the ϵ -floor), $d_{\text{eff}}(\Sigma_G) \leq r + \mathcal{O}(\epsilon)$, because $\mathcal{O}_{G,r}$ has at most r nonzero eigenvalues. Spectral truncation thus caps the number of collective angle directions the query policy explores; table 4 reports d_{eff} increasing monotonically with r .*

Proposition 5 (Query-budget tradeoff under coverage). *Adopt a coverage model in which the refinement of section 3.3 reaches the ε -optimal subset of the trust region once it has probed a number of collective directions proportional to $d_{\text{eff}}(\Sigma_G)$, provided those directions retain the ε -optimal mass. Writing $m(r)$ for the fraction of near-optimal mass captured by the leading r operator directions, the expected queries to reach a target ratio τ obey*

$$\mathbb{E}[Q_G(\tau)] \lesssim \frac{c d_{\text{eff}}(\Sigma_G^{(r)})}{m(r)} \quad (11)$$

for a constant c independent of r . The numerator decreases with truncation (proposition 4) while $m(r)$ is nondecreasing in r ; the minimiser r^* is therefore interior whenever aggressive truncation sheds near-optimal mass faster than it concentrates the search. This is the query-budget analogue of the representation-versus-complexity tradeoff of spectral-truncation kernels⁵: there the truncation parameter trades approximation against generalisation; here it trades near-optimal coverage against the number of objective queries.

The bound eq. (11) is a scaling statement, not a sharp constant; it is instantiated in table 1 (mean queries-to-target $\bar{Q}_{0.98}$) and table 4 (d_{eff} and the interior optimum r^*).

4 Experiments

The manuscript configuration is $p = 3$, $Q = 24$, truncation rank $r = 4$, commutator weight $\omega = 4.0$, and seed 260424803. The training library contains six graphs per family from {Erdos–Renyi, random-regular, Watts–Strogatz, Barabasi–Albert} with $n \in \{8, 10\}$. The test set contains four held-out graphs per family, for 16 paired instances. All objectives are exact statevector expectations and all MaxCut normalizers are exact brute-force optima.

The baselines are:

- **Random:** uniform random angles under the same query budget.
- **TQA:** one fixed trotterized quantum annealing schedule.
- **TQA+coordinate:** TQA anchor followed by isotropic coordinate refinement.
- **kNN+coordinate:** feature-neighbour posterior mean with diagonal coordinate refinement.
- **OST diagonal:** the same operator posterior as OST-QAOA but with off-diagonal covariance removed.
- **OST-QAOA:** full operator covariance and eigendirection search.

5 Results

OST-QAOA is the strongest method in the matched-budget benchmark (table 1 and fig. 2). It obtains 0.818 ± 0.018 mean ratio, $+0.110$ over TQA+coordinate with a paired bootstrap interval $[+0.080, +0.141]$, and wins on all 16 paired test graphs. The diagonal operator variant is also strong at 0.806 ± 0.020 , but the full covariance improves the mean by 0.012, showing that the off-diagonal operator geometry is measurable in this configuration.

Beyond the final ratio, the operator prior is markedly more *query-efficient*: OST-QAOA reaches 98% of the best observed ratio in a mean of 11.6 objective queries, against 25.0 for

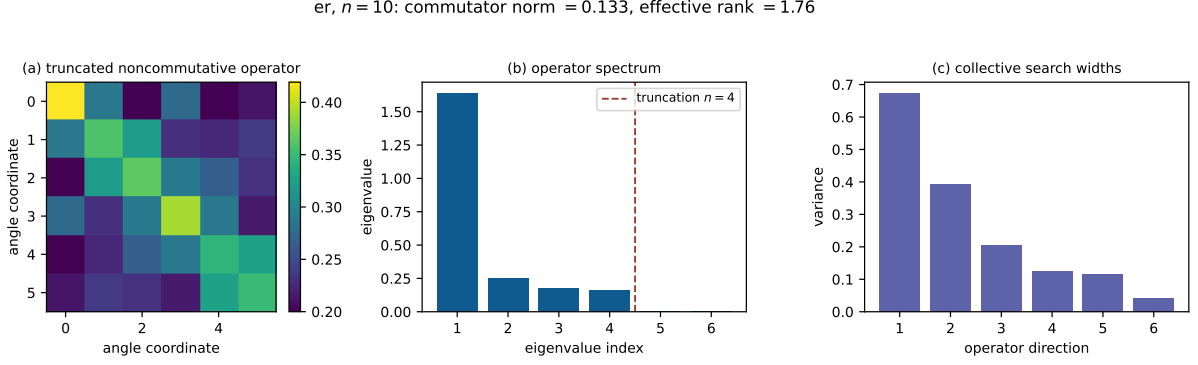


Figure 1. Operator-spectral prior. A held-out graph is mapped to a truncated $2p \times 2p$ angle operator, its retained spectrum, and the resulting posterior search widths. The rank parameter controls how many noncommutative graph directions can influence the query policy.

Table 1. Matched-budget benchmark at $p = 3$, $Q = 24$, rank 4. The table is generated by `uqqaoa-reproduce`; values are mean approximation ratio with 95% confidence intervals across 16 held-out graphs. Δ is the paired difference against TQA+coordinate with a bootstrap 95% interval; $\bar{Q}_{0.98}$ is the mean number of objective queries to reach 98% of the best observed ratio (proposition 5).

Method	Mean ratio	Δ vs. TQA+coord. [95% CI]	Wins	Sign p	$\bar{Q}_{0.98}$
Random	0.743 ± 0.028	+0.035 [+0.001, +0.070]	11/16	0.2101	24.1
TQA	0.614 ± 0.030	-0.093 [-0.156, -0.040]	0/16	$< 10^{-4}$	25.0
TQA+coordinate	0.708 ± 0.038	0.000	0/0	1.0000	25.0
kNN+coordinate	0.740 ± 0.042	+0.032 [+0.012, +0.056]	13/16	0.0213	23.6
OST diagonal	0.806 ± 0.020	+0.099 [+0.067, +0.130]	15/16	0.0005	14.4
OST-QAOA	0.818 ± 0.018	+0.110 [+0.080, +0.141]	16/16	$< 10^{-4}$	11.6

TQA+coordinate and 14.4 for the diagonal operator variant ($\bar{Q}_{0.98}$, table 1)—a roughly $2.2\times$ reduction in queries to comparable quality, which is the operationally relevant quantity in proposition 5.

The ablation in table 3 and fig. 3 attributes the improvement. Restricting the covariance to its diagonal—the commutative special case of proposition 3—drops the mean ratio to 0.806 ± 0.020 (-0.012), so the off-diagonal, noncommuting search directions are operationally used. The truncation sweep (fig. 3 and table 4) shows the interior optimum predicted by proposition 5: rank $n^*=4$ is best, aggressive truncation ($n=1$) underfits the angle geometry, and no truncation ($n=2p=6$) over-diffuses the covariance; the effective dimension d_{eff} rises monotonically from 2.50 at $n=1$ to 3.65 at $n=6$, confirming proposition 4. The explicit squared-commutator energy term is, by contrast, near-neutral at the operating point (+0.001 over commutator-off in the sweep); we report this honestly—the operationally relevant noncommutativity already resides in the truncated $\mathcal{A}_G, \mathcal{B}_G$ geometry and the eigenbasis search, not in the squared-commutator energy.

Figure 4 shows that the gain is not only a final-query artifact. The operator mean and nearest-neighbour anchor improve early incumbent quality, while covariance eigendirections preserve gains as the budget is spent.

6 Reproducibility and Package Interface

The manuscript is backed by an installable package in `submission/code`. From that directory,

```
pip install .
uqqaoa-reproduce --output-dir . \
```

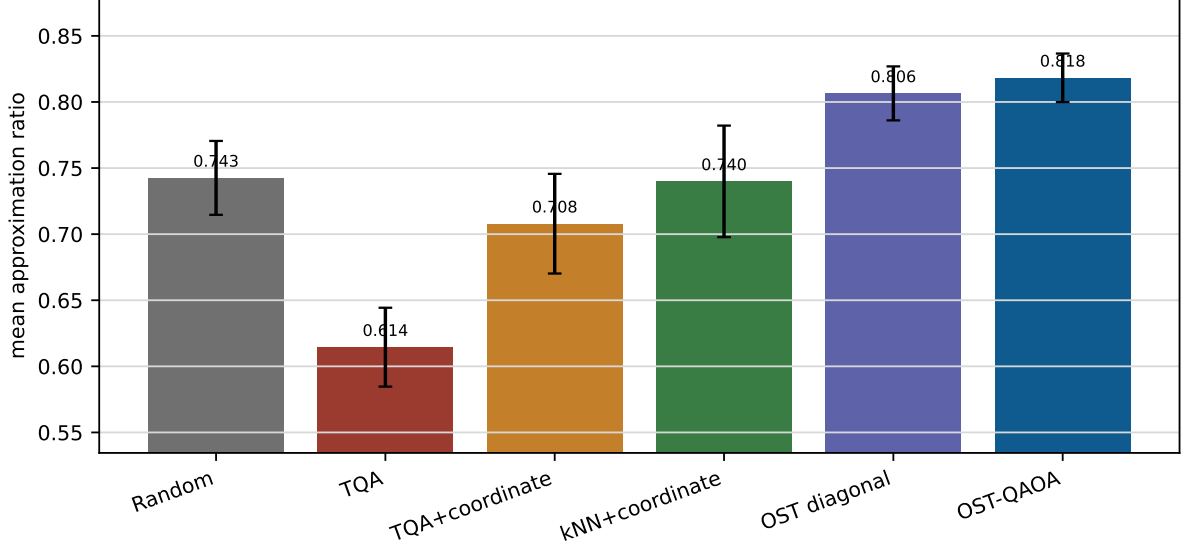


Figure 2. Mean approximation ratio by method. OST-QAOA and the diagonal operator ablation outperform TQA+coordinate and kNN+coordinate under the same objective-query budget.

Table 2. Per-family paired breakdown. Each family contributes four held-out graphs. The small per-family sample size makes family-level sign tests conservative, but every family has a positive paired mean delta.

Family	N	TQA+coord.	OST-QAOA	Δ	Wins	Sign p
er	4	0.675	0.789	+0.114	4/4	0.1250
random_regular	4	0.818	0.847	+0.029	4/4	0.1250
watts_strogatz	4	0.705	0.814	+0.109	4/4	0.1250
barabasi_albert	4	0.633	0.822	+0.189	4/4	0.1250

```
--depth 3 --budget 24 --rank 4 \
--commutator-weight 4.0
```

regenerates the manuscript figures, tables, CSV summaries, query traces, and metadata. A fast end-to-end sanity check is

```
uqqaoa-reproduce --output-dir /tmp/uq-qaoa-smoke --smoke
```

The same functionality is available through the Python API:

```
from uq_qaoa import build_operator_library, ost_qaoa_search
from uq_qaoa.graphs import generate_graph
```

```
library = build_operator_library(
    p=3, rank=4, commutator_weight=4.0, train_per_family=6
)
graph = generate_graph("random_regular", 10, 1234)
result = ost_qaoa_search(graph, library, budget=24)
```

7 Discussion

Relationship to spectral-truncation kernels. Hashimoto et al.⁵ introduce positive-definite, noncommutative C^* -algebraic kernels whose truncation parameter trades representation power against model complexity, validated on supervised vector- and function-valued

Table 3. Design ablation. The full row is the same rank-4 OST-QAOA configuration used in table 1. Replacing the covariance by its diagonal removes the directional operator advantage; disabling the commutator term is nearly neutral at the operating point but remains a controlled part of the spectral-truncation sweep.

Variant	Mean ratio	Δ vs. full
OST-QAOA (full)	0.818 ± 0.018	+0.000
diagonal directions	0.806 ± 0.020	−0.012
commutator off	0.817 ± 0.023	−0.001
TQA + coordinate	0.708 ± 0.038	−0.110

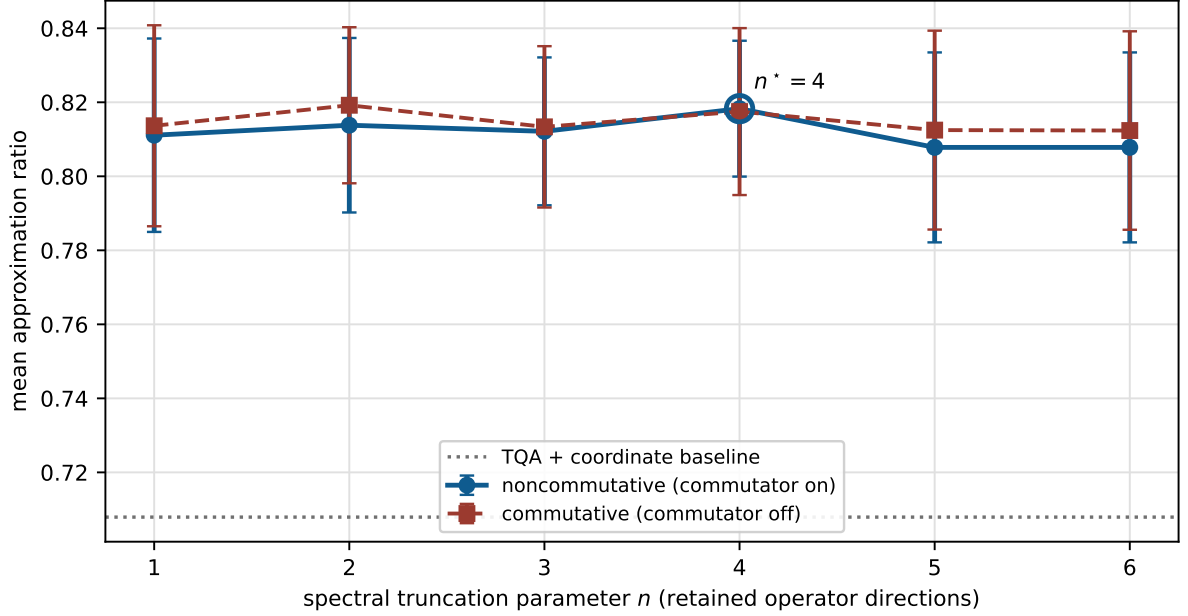


Figure 3. Spectral-truncation sweep (the central experiment). Mean approximation ratio versus the truncation parameter $n \in \{1, \dots, 2p\}$, with the noncommuting operator (commutator on) and its commutator-off counterpart, against the TQA+coordinate baseline. The curve has an interior optimum n^* : aggressive truncation ($n=1$) and no truncation ($n=2p$) are both worse, the query-budget analogue of the representation–complexity tradeoff of spectral-truncation kernels⁵. Effective dimension increases monotonically with n (table 4, proposition 4).

regression. OST-QAOA carries the same design pressure—finite-rank noncommutativity as a tractable way to encode interactions—into a setting their framework does not address, and advances it on three axes. (i) *New domain.* The truncated noncommuting operator is not a regression kernel but a graph-conditioned prior over variational-quantum parameters, converted into a deterministic, auditable objective-query policy. (ii) *An operational bound.* Where their tradeoff is statistical (a Rademacher generalisation bound), ours is operational: proposition 5 ties the truncation parameter to the *number of objective queries* through the effective dimension $d_{\text{eff}}(n)$, the quantity that actually limits near-term QAOA, and the predicted interior optimum n^* is observed (fig. 3). (iii) *A demonstrated mechanism.* The diagonal (commutative) restriction of proposition 3 is measurably worse than the full noncommuting geometry, so the value of operating off the coordinate axes is exhibited, not assumed. These are the senses in which the present work reaches and extends the noncommutative-truncation idea rather than merely citing it.

The present experiments are intentionally audit-sized. They use small exact statevectors and brute-force MaxCut optima so that every number can be regenerated on a CPU. This

Table 4. Spectral-truncation sweep. Mean approximation ratio at each truncation parameter n with the noncommuting operator (on) and its commutator-off counterpart (off), Δ_{nc} their difference, and the effective search dimension d_{eff} (definition 2). The starred row n^* is the sweep optimum; d_{eff} rises monotonically with n , instantiating proposition 4.

Truncation n	Noncomm. (on)	Comm. (off)	Δ_{nc}	d_{eff}
1	0.811	0.814	-0.003	2.50
2	0.814	0.819	-0.005	2.93
3	0.812	0.813	-0.001	3.24
4*	0.818	0.817	+0.001	3.46
5	0.808	0.812	-0.005	3.65
6	0.808	0.812	-0.005	3.65

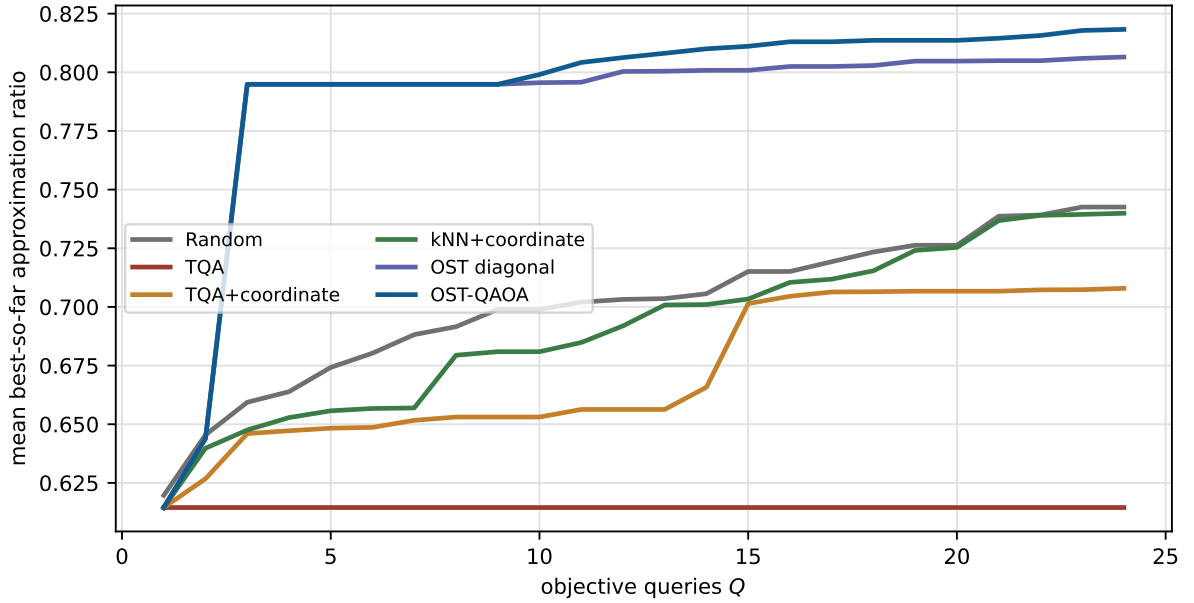


Figure 4. Best-so-far query curves. The package records every objective evaluation; the plot averages incumbent ratios by query index. Operator anchors improve the early budget, and eigendirection refinement widens the gap after the initial anchors.

is appropriate for a submission artifact, but it is not a claim of hardware-scale biomedical deployment. The next experimental step is to move from unweighted synthetic benchmark families to weighted biomedical graphs derived from transcriptomic, drug-target, and pathway data, then evaluate finite-shot robustness and hardware noise.

8 Conclusion

OST-QAOA replaces a diagonal uncertainty heuristic with a rank-controlled, noncommuting operator prior on QAOA angle space. The method uses the off-diagonal operator geometry to choose collective search directions, ships as an installable Python package, and regenerates every manuscript result from tunable parameters. In the exact-statevector benchmark the operator-spectral policy outperforms budget-matched TQA coordinate refinement on all 16/16 paired test graphs (+0.110, $p < 10^{-4}$) and reaches comparable quality in roughly half the queries, while the truncation sweep exhibits the interior optimum predicted by the query-budget tradeoff (proposition 5) and the ablation localises the advantage to the spectral-truncation parameter and the off-diagonal operator directions rather than to the squared-commutator energy.

Table 5. Artifact manifest. Each manuscript display item is generated by the package-native artifact driver.

Element	Generator	Output
Fig. 1	plot_operator_spectrum	fig01_operator_spectrum.pdf
Fig. 2	plot_benchmark	fig02_benchmark.pdf
Fig. 3	plot_truncation_sweep	fig03_truncation_sweep.pdf
Fig. 4	plot_query_curves	fig04_query_curves.pdf
Table 1	summarize_results	table01_headline.tex
Table 2	run_ablation	table02_ablation.tex
Table 3	run_truncation_sweep	table05_truncation.tex
Table 4	summarize_per_family	table04_per_family.tex

Acknowledgments

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Data and Code Availability

All code required to regenerate the results is in `submission/code`. The package command `uqqaoa-reproduce` writes the figures, tables, CSV files, and query traces used by this manuscript.

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